



Instructor: Date:

Learner: Regional Branch:

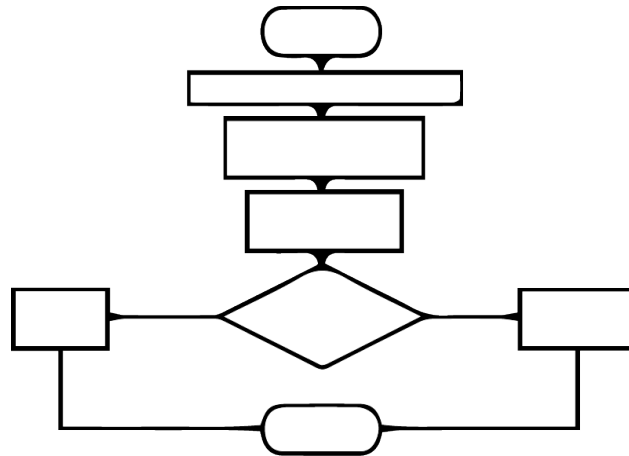
Program: Center:

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**Let's Start!**

The Importance of Flow Diagrams in Data Processing.

**Learning Outcome:**

Describe common activities at work by using short sentences in familiar spoken and written situations.

Interpret the requirements report to determine the technological needs in managing information, based on rules and protocols established by the company.

Can Do Statements:

- I can talk about different facts related to data flow diagrams.
- I can give information related to the history of data flow diagrams.
- I can explain the importance of flow diagrams.

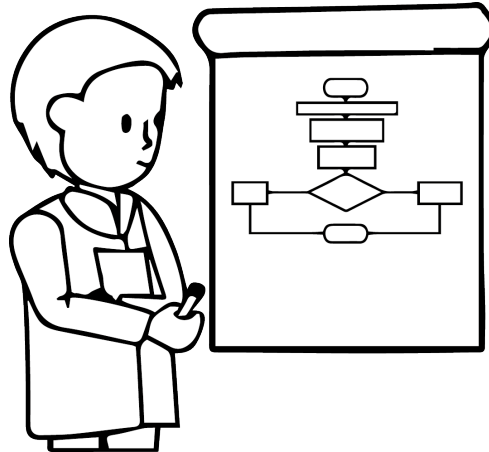
Ask your partner:

- How can you show or represent a process easily? Give some examples.
- Where can you find flow diagrams?



Let's Explore!

We require a new *DFD.



Instructions: Read the text below.

Technician: I just completed a full analysis of the information system that the company had 2 years ago. So, last month I found serious problems with some processes in the company.

Operator: What do you mean?

Technician: First, the company has a lot of information for each process. It is necessary to organize the data. If we want to represent processes in an efficient way, we need to use data flow diagrams. I think we developed one 5 years ago.

Operator: Sorry, I don't know what that is.

Technician: A data flow diagram, or DFD, is a graphic representation of the flow of data, with the support of an information system.

Operator: Ok, that is something new.

Technician: It is not! Four decades ago, Larry Constantine proposed the concept of DFD. For its development, he took the Data Flow Graph model of computation. In the year 1970, Data Flow Diagrams became a systematic and popular way to see how a project evolves step by step. A few years later, people used the model to present data flow in computers. That was a very important fact for software development.

Operator: I see. So, do you think we need to design a new DFD?

Technician: That seems to be the solution. If we develop a data flow diagram, we know what kind of information is input to and output from the system. This is important because it also shows the stored data and provides a visualization of data processing. However, if you want information about the timing of the process or the sequence, the DFD doesn't show these elements. But, we can use an ADDIE model.

Operator: This seems to be useful for organizing information. Besides, it shows the interaction between elements, so that if we develop a flow diagram, we can structure and order this complex system.

**Let's Practice!**

We are going to organize the input the output information of the company.

Instructions: Use the words from the word bank to complete each sentence.

need to / want to visualize / don't know,
doesn't know / needs to / wants to visualizing

1. If we _____ systematize information in our company, we need to use data flow diagrams.
2. If we _____ data processing, we can use a data flow diagram.
3. If you _____ the input and the output of your data system, you can also use the data flow diagram.

**Let's Practice!**

What does data flow diagrams show?

Instructions: Select the correct answer in order to complete the sentence.

1. _____, we know what kind of information is input to and output from the system.
 - a. If we develop a data flow diagram.
 - b. If we don't develop a data flow diagram.
 - c. If we avoid using a data flow diagram.
 - d. If we developed a data flow diagram.
2. If we implement the DFD, _____.
 - a. we don't provide a visualization of data processing.
 - b. we get a visualization of data processing.
 - c. it provides a visualization of data processing.
 - d. we can't get a visualization of data processing.
3. If you need the information about the timing of the process or the sequence, _____. (HINT: Pay attention to the structure of these sentences. Do we need a future or a present tense?)
 - a. use the DFD, which is the only tool you have to see these elements.
 - b. you won't need to use a DFD.
 - c. don't use the DFD since it doesn't show these elements.
 - d. you will need to use a DFD.
4. We can structure and order complex information _____.
 - a. if you don't know how to use a flow diagram.
 - b. and we develop a flow diagram.
 - c. if we don't develop a flow diagram.
 - d. if we develop a flow diagram.
5. _____, you need to use a data flow diagram.
 - a. If you want to represent how data flows.
 - b. If you don't want to represent how data flowed.
 - c. When you can't represent how data flows.
 - d. If you will represent how data flows.



**Extension Activity**

Let's find out information about data flow diagrams.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Read about DFD.
- Write down on your notebook its origin, definition and some important facts about it.
- Draw an example of a DFD in ADSI.
- Present your work to your classmates and check pronunciation with your instructor.

Self-Assessment**Yes****No**

- I talk about different facts related to data flow diagrams.
- I give information related to the history of data flow diagrams.
- I explain the importance of flow diagrams.

☐ ☐☐ ☐☐ ☐**Target Words**

- flow** (noun): a continuous movement of something
e.g. Besides, it shows the interaction between elements, so that, if we develop a flow diagram, we structure and order this complex system.
- develop** (verb): to grow or improve
e.g. I think we developed an analysis of the information system 5 years ago.
- timing** (noun): the control of the speed in a process or event
e.g. However, if you want information about the timing of the process or the sequence, the DFD doesn't show these elements.
- input** (noun): the data that enters a computer or a machine for a process
e.g. If we develop a data flow diagram, we know what kind of information is input to and output from the system.
- output** (noun): the result of a process
e.g. If we develop a data flow diagram, we know what information belongs to the input and what to the output in the system.
- store** (verb): to collect or copy information to keep it
e.g. This is important because it also shows the data stored and provides a visualization of data processing.



Learning Points

Zero Conditional In Affirmative And Negative.

You use the zero conditional to talk about real situations and facts. You can use the zero conditional in both negative and affirmative statements.

You use the next structure to explain two situations that complement each other and that are always real.

Structure: **If + Present Simple, Present Simple**

For example:

*If we **want** to represent processes in an efficient way, we **use** the data flow diagrams.*

The first part explains a **condition** and the second one explains the **result**, consequence or direct relation to the first condition. Both parts need to be in present simple and they are always about true situations that are obvious or general.

One or both of the parts of the sentence can be a negative statement. Remember to use the auxiliary verbs when necessary.

Structure: **If + Present Simple (negative), + Present Simple (negative)**.

For example:

***If you don't use** a graphical representation of how the data flows, **you don't get** a general sketch of the system.*

Notice that the clauses are separated by a comma; when the sentence starts with the phrase expressing the consequence, do not use the comma to separate them. Look at the example:

***You get** a general sketch of the system **if you use** a graphical representation of how the data flows.*

You can also replace **If** by **when** without changing the meaning of the sentence.

Examples

- *If we develop a data flow diagram, we know what kind of information is input to and output from the system.*
- *If you want information about the timing of the process or the sequence, the DFD doesn't show these elements.*
- *If we develop a flow diagram, we can structure and order this complex system.*





Learning Points

The Addie Model.

You can use an Addie model to create an effective training and performance of a process.

In an ADDIE framework, you can find 5 phases of a generic process. Instructional designers and training developers use this model to have a guideline for effective performance. The objective is to have a constant revision or feedback of the specific process. The ADDIE model is dynamic and interactive. Let's take a look at the description of its phases.

- **Analysis:** In this phase, the designer identifies the objectives and possible problems. Also, he needs to analyze the environment, the advantages, and disadvantages.
- **Design:** The instructional designer plans strategies to fulfill the objectives. This phase needs to be specific and systematic, this means it has to be organized and logical.
- **Development:** It is the moment to create and develop the designs. The developers can include more elements according to the analysis and design. Also, they can revise and give feedback to the process.
- **Implementation:** In this phase, the designers apply the procedures and elements developed. At the same time, they test and analyze its application.
- **Evaluation:** The evaluation of the process is the most important element. Remember you can find this phase in the others; it can be formative and summative.